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Equations of State — Introduction

Motivation

Over the next several lectures we will explore another major component of stellar microphysics: the **equation of state (EOS)**. In general, the EOS specifies the pressure as a function of density, temperature, and composition:

$$P = P(\rho, T, X_i)$$

Its most direct appearance in stellar structure is through the equation of hydrostatic equilibrium:

$$\frac{dP}{dr} = -\rho \frac{GM(r)}{r^2}$$

But the EOS also determines important microscopic properties of stellar matter, such as the ionization fraction, which strongly affects energy transport.

Equilibrium Assumptions

Throughout this discussion we assume local equilibrium, including:

- Pressure equilibrium
- Thermal equilibrium
- Ionization equilibrium

We explicitly *do not* assume nuclear equilibrium, since that would preclude fusion. Nuclear reactions are typically slow compared to microscopic equilibration processes.

The following discussions of EoS are made using an assumption that the radiation, gas, solid, material, is in a state of local thermodynamic equilibrium (LTE).

Some definitions:

The pressure scale height is defined as

$$\lambda_P = \left(\frac{\text{\textcolor{red}{d}ln} P}{dr} \right)^{-1} \equiv \frac{P}{g\rho}$$

where we used the equation of HSE to simplify. Also, sometimes the pressure scale height is defined as H_P with value of order unity in the constant-density model.

Consider also the mean free path λ_{phot} of a photon, or the distance traveled by a photon before it is either absorbed or scattered in a new direction:

photon mean free path

$$\lambda_{\text{phot}} = (\kappa\rho)^{-1} \text{ (cm)}$$

We can use an estimate with the electron opacity of $\kappa \approx 1 \text{ cm}^2 \text{ g}^{-1}$ and average density of about 1 g cm^{-3} , we find that λ_{phot} is about a centimeter and that λ_P is about $\sim 10^{11}$, several orders of magnitude larger.

In-class exercise: Pressure Scale Height

If we assume:

- Ideal gas pressure:

$$P = \frac{\rho k T}{\mu m_p}$$

- Gravitational acceleration:

$$g \sim \frac{GM}{R^2}$$

Tasks

1. Show that the pressure scale height scales as

$$\lambda_P \sim \frac{kT}{\mu m_p g}.$$

2. Estimate λ_P near the **solar photosphere**, using:

- $T \approx 5800 \text{ K}$
- $\mu \approx 0.6$
- $g_{\odot} \approx 2.74 \times 10^4 \text{ cm s}^{-2}$

3. Compute λ_P numerically in Python.

4. Compare λ_P to the solar radius $R_{\odot} = 6.96 \times 10^{10} \text{ cm}$ and interpret the result physically.

► Solution

Distribution Functions

The *distribution function* for a species of particle measures the number density of that species in the combined 6-dimensional space of coordinates plus momenta.

Some definitions:

The relation between the number density of some species of elementary nature (ions, photons, etc.) in coordinate-momentum space and its chemical potential in thermodynamic equilibrium is found from statistical mechanics to be

distribution function

$$n(p) = \frac{1}{h^3} \sum_j \frac{g_j}{\exp\{[-\mu + \epsilon_j + \epsilon(p)]/kT\} \pm 1}$$

where

1. μ - chemical potential of the species
2. j - refers to the possible energy states of the species (e.g., energy levels of an ion)
3. ϵ_j - is the energy of state j referred to some reference level
4. g_j - is the energy of state j referred to some reference level
5. $\epsilon(p)$ - is the kinetic energy as a function of momentum p
6. $+$ - for Fermi-Dirac particles (fermions of half-integer spin)
7. $-$ - for Bose-Einstein particles (bosons of zero or whole integer spin)
8. h - Planck's constant

Compressibility: Gas vs Liquid

Everyday examples help illustrate equations of state:

- Air is easily compressed: pressure increases slowly with density.
- Water is nearly incompressible: pressure increases rapidly with density.

An ideal gas follows:

$$P = nkT$$

whereas liquids behave very differently, forming the basis of hydraulic systems.

Chemical Potential

Our thermodynamic starting point is the chemical potential:

$$\mu_i = \left(\frac{\partial E}{\partial N_i} \right)_{S,V}$$

For chemical equilibrium involving reactions that change particle numbers:

$$\sum_i \mu_i dN_i = 0$$

For photons, particle number is not conserved, so:

$$\mu_\gamma = 0$$

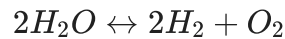
Stoichiometry and Equilibrium

Because reactions link particle numbers, we can also write:

$$\sum_i \mu_i \nu_i = 0$$

Example: Water formation

For



equilibrium implies:

$$2\mu_{H_2O} = 2\mu_{H_2} + \mu_{O_2}$$

Quantum Statistical Mechanics

Consider particles in equilibrium at temperature T with chemical potential μ . The phase-space distribution function is:

$$n(p) = \frac{1}{h^3} \sum_{\text{states}} \frac{1}{e^{E_{\text{tot}}/kT} \pm 1}$$

where:

$$E_{\text{tot}} = -\mu + E(\text{state}) + E(p)$$

- + for fermions
 - - for bosons
-

Number Density, Pressure, and Energy Density

Assuming isotropy:

$$n = \int n(p) 4\pi p^2 dp$$

$$P = \frac{1}{3} \int n(p) p v 4\pi p^2 dp$$

$$u = \int n(p) E(p) 4\pi p^2 dp$$

with:

$$E(p) = \sqrt{p^2 c^2 + m^2 c^4} - m c^2$$

Blackbody Radiation

Photons are massless bosons of unit spin. Since they travel at c , they only have two states (two spin orientations or polarizations) for a given energy and thus the degeneracy factor in (3.8) is $g = 2$.

We can define a radiation pressure P_{rad}

$$P_{\text{rad}} = \frac{a T^4}{3} \text{ (dyne cm}^{-2}\text{)}$$

and the energy density for a radiation dominated gas

$$E_{\text{rad}} = a T^4 \equiv 3 P_{\text{rad}} \text{ (erg cm}^{-3}\text{)}$$

where a is the radiation constant $7.566 \times 10^{-15} \text{ (erg cm}^{-3} \text{ K}^{-4}\text{)}$. We note that this is the result for a "gamma-law" EoS for $\gamma = 4/3$ from Eqn (1.22).

We can also define the frequency dependent Planck function:

$$B_\nu(T) = \frac{c}{4\pi} u_\nu$$

and the integrated value over all wavelengths:

$$B(T) = \frac{\sigma}{\pi} T^4$$

Integrating the functional form of this equation shows the shape of the radiation curve:



A plot of the function $B(x) = \frac{x^3}{[e^x - 1]}$ corresponding to the vital part of either uv or the Planck function. The maximum is at $x = h\nu/kT = 2.821$. If we use the central temperature of the Sun of $T_c \approx 10^7 \text{ K}$ this would imply a photon energy of 2.4 keV.

For photons:

- $\mu = 0$
- $g = 2$
- $E(p) = pc$

The photon number density is:

$$n_\gamma = \frac{8\pi}{h^3} \int_0^\infty \frac{p^2 dp}{e^{pc/kT} - 1} \approx 20 T^3 \text{ cm}^{-3}$$

Radiation pressure and energy density are:

$$P_{\text{rad}} = \frac{1}{3} a T^4$$

$$u_{\text{rad}} = a T^4$$

where:

$$a = 7.6 \times 10^{-15} \text{ erg cm}^{-3} \text{ K}^{-4}$$

In-class exercise: Blackbody Radiation and the Planck Spectrum

Photons are massless bosons with chemical potential $\mu_\gamma = 0$.

Their equilibrium distribution leads to the **Planck spectrum** for blackbody radiation.

The frequency-dependent Planck function is

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1}.$$

Tasks

1. Define the dimensionless variable

$$x \equiv \frac{h\nu}{kT}$$

and show that the *shape* of the Planck spectrum can be written as

$$B(x) \propto \frac{x^3}{e^x - 1}.$$

2. Using Python, plot $B(x)$ for $0 < x < 10$.
3. Numerically identify the value of x where $B(x)$ is maximized.
4. Using your result, explain why the peak photon energy scales linearly with temperature.

► Solution

Monatomic Ideal Gas

We start by assuming that the gas has $\mu/kT \ll -1$. This allows us to plot the relevant portion of the distribution for a Maxwell-Boltzmann Ideal Gas ($C(x) = \sqrt{x}e^{-x}$):



A plot of the function corresponding to the exciting part of the Maxwell-Boltzmann distribution in energy space. The maximum is at $x = 1/2$ ($\epsilon = kT/2$).

After some substitutions we can define the pressure and internal energy for an ideal gas:

$$P_{\text{ideal}} = nkT \text{ (dyne cm}^{-2}\text{)}$$

$$E_{\text{ideal}} = \frac{3}{2}nkT \text{ (erg cm}^{-3}\text{)}$$

For a sparse, nonrelativistic, monatomic gas:

$$E(p) = \frac{p^2}{2m}, \quad v = \frac{p}{m}$$

The chemical potential satisfies:

$$e^{\mu/kT} = \frac{nh^3}{g(2\pi mkT)^{3/2}}$$

and we recover:

$$P = nkT$$

$$u = \frac{3}{2}nkT$$

In-class Question

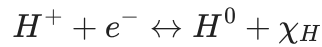
Why is the chemical potential of an ideal gas very negative?

► Solution

Saha Equation (Thermal Ionization)

an expression that relates the ionization state of a gas in thermal equilibrium to the temperature and pressure

Consider:



with $\chi_H = 13.6$ eV.

Number densities:

$$n_e = \frac{2(2\pi m_e kT)^{3/2}}{h^3} e^{\mu^-/kT}$$

$$n^+ = \frac{(2\pi m_p kT)^{3/2}}{h^3} e^{\mu^+/kT}$$

$$n^0 = \frac{2(2\pi m_p kT)^{3/2}}{h^3} e^{\mu^0/kT} e^{\chi_H/kT}$$

Equilibrium requires:

$$\mu^- + \mu^+ - \mu^0 = 0$$

yielding the **Saha equation**:

$$\frac{n^+ n_e}{n^0} = \left(\frac{2\pi m_e kT}{h^2} \right)^{3/2} e^{-\chi_H/kT} \approx 2.415 \times 10^{15} T^{3/2} e^{-\chi_H/kT} (\text{cm}^{-3})$$

for the single-level pure hydrogen gas. We can redefine the degree of ionization,

$$y = \frac{n^+}{n} \equiv \frac{n_e}{n}$$

Take away - in practice, semi-analytic methods are not accurate and you are left with using tables.

Ionization Fraction

Define:

$$y \equiv \frac{n^+}{n}$$

Then:

$$\frac{y^2}{1-y} = \frac{4 \times 10^{-9}}{\rho} T^{3/2} e^{-1.6 \times 10^5/T}$$

In-class Question

Why does the ionization fraction increase as density decreases at fixed temperature?

► Solution

Limits of Applicability

Applying the Saha equation to the solar core gives a misleading result: partial neutrality. The resolution is pressure ionization, which dominates at high densities.

Another issue arises from summing over *all* atomic energy states, which formally diverges.

In-class Question

What physical effect truncates the atomic level structure in real stellar plasmas?

► Solution

In-class Question

At fixed density, why does the ionization transition occur over a relatively narrow temperature range?

► Solution

In-class Exercise: Hydrogen Ionization with the Saha Equation

In this exercise, we explore how the ionization fraction of hydrogen depends on temperature and density using the Saha equation.

Part A: Conceptual Questions

1. At fixed temperature, how do you expect the ionization fraction to change as density decreases?
2. At fixed density, why does ionization occur over a narrow temperature range?

Part B: Python Exploration

Complete the code below to compute and plot the ionization fraction $y(T)$ for several fixed densities.

Part C: Interpretation

1. For which density does hydrogen ionize at the lowest temperature?
2. Why are stellar atmospheres easier to ionize than stellar interiors?
3. How does this behavior relate to the assumptions behind the Saha equation?

► Solution

Equations of State: Degeneracy and Interactions

These EoS describe electrons, protons, and neutrons. The prime motivation for this discussion is that the equation of state in the inner regions of many highly evolved stars, including white dwarfs, is dominated by degenerate electrons and, to a great extent, this determines the structure of such stars.

The Completely Degenerate Gas

Assume that the gas is at zero temperature. The critical kinetic energy at which the Fermi energy is reached is defined as the Fermi energy ϵ_F .

 e_fermi

The function $F(\epsilon/\epsilon_F)$ from (3.47) is plotted versus particle kinetic energy in units of the Fermi energy for $T = 0$. Fermions occupy all states up to ϵ_F , corresponding to the shaded area of unit height and width. No particles have energies greater than ϵ_F .

The dashed line shows how $F(\epsilon)$ changes when the temperature is raised slightly; in this example, $\epsilon_F/kT = 20$.

Note: This is a rough conceptual illustration. In reality, there is no sharp boundary on the T - ρ - μ_e plane separating degenerate from nondegenerate electrons.

Fermi–Dirac Particles

Consider a particle with spin 1/2 (electron, proton, neutron). These are fermions, so their occupation numbers obey Fermi–Dirac statistics. Choosing the reference energy to be mc^2 , the number density is

$$n = \frac{8\pi}{h^3} \int_0^\infty \frac{p^2 dp}{e^{[-\mu + mc^2 + E(p)]/kT} + 1}$$

where the kinetic energy is

$$E(p) = \sqrt{m^2 c^4 + p^2 c^2} - mc^2$$

and the velocity is

$$v(p) = \frac{\partial E}{\partial p} = \frac{p}{m} \left[1 + \left(\frac{p}{mc} \right)^2 \right]^{-1/2}$$

Consistency checks

- $p \ll mc \Rightarrow v \approx p/m$ (nonrelativistic)
- $p \gg mc \Rightarrow v \rightarrow c$ (ultrarelativistic)

The transition occurs near $p \sim mc$.

Completely Degenerate Gas

Degeneracy occurs when particles are forced to occupy higher momentum states due to the Pauli exclusion principle. In the limit

$$kT \ll E - (\mu - mc^2)$$

the occupation function becomes

$$F(E) = \frac{1}{e^{[E - (\mu - mc^2)]/kT} + 1}$$

In-class Question

What happens to $F(E)$ as $kT \rightarrow 0$ when the exponent is positive or negative?

► Solution

Fermi Momentum and Fermi Energy

Using the uncertainty principle,

$$p_F \sim \hbar n^{1/3}$$

Define the relativity parameter

$$x_F \equiv \frac{p_F}{mc}$$

Then

$$E_F = mc^2 \left(\sqrt{1 + x_F^2} - 1 \right)$$

The chemical potential is

$$\mu_F = E_F + mc^2$$

Number Density of a Completely Degenerate Fermion Gas

How does the number density integral simplify at $kT \rightarrow 0$?

Since all states up to p_F are occupied,

$$n = \frac{8\pi}{h^3} \int_0^{p_F} p^2 dp = \frac{8\pi}{3} \left(\frac{h}{mc} \right)^{-3} x_F^3$$

Numerically,

$$n = 5.9 \times 10^{29} \left(\frac{m}{m_e} \right)^3 x^3 \text{ cm}^{-3}$$

For electrons,

$$\frac{\rho}{\mu_e} \approx 10^6 x^3 \text{ g cm}^{-3}$$

Electrons become relativistic when $\rho \gtrsim 10^6 \text{ g cm}^{-3}$.

Scaling of the Fermi Energy

- Nonrelativistic:

$$E_F \sim \rho^{2/3}$$

- Ultrarelativistic:

$$E_F \sim \rho^{1/3}$$

Since $m_e c^2 = 511 \text{ keV} \approx 6 \times 10^9 \text{ K}$, degeneracy becomes important at surprisingly low temperatures.

Fermi Pressure and Energy Density

For a strongly degenerate electron gas,

$$P_e = \frac{\pi}{3} \left(\frac{h}{m_e c} \right)^{-3} m_e c^2 f(x) \equiv A f(x)$$
$$E_e = A g(x)$$

where

$$f(x) = x(2x^2 - 3)\sqrt{1 + x^2} + 3 \sinh^{-1} x$$

$$g(x) = 8x^3(\sqrt{1 + x^2} - 1) - f(x)$$

Limiting behavior:

$$f(x) \rightarrow \begin{cases} \frac{8}{5}x^5, & x \ll 1 \\ 2x^4, & x \gg 1 \end{cases}$$

In-class Question

Which particles become degenerate first, and why?

► Solution

Where is the astrophysics?

White Dwarfs and Degeneracy Pressure

White dwarfs are supported by electron degeneracy pressure, not thermal pressure.

In the nonrelativistic regime:

$$E_F \sim \frac{\hbar^2 N^{2/3}}{2m_e R^2}$$

Gravitational energy per electron:

$$E_G \sim -\frac{GNm_B^2}{R}$$

Total energy per electron:

$$E_{\text{tot}} = \frac{\hbar^2 N^{2/3}}{2m_e R^2} - \frac{GNm_B^2}{R}$$

Minimizing yields:

$$R \sim M^{-1/3}$$

Relativistic Degeneracy and the Chandrasekhar Mass

In the relativistic limit,

$$E_F \sim \frac{\hbar c N^{1/3}}{R}$$

so

$$E = \frac{1}{R} \left(\hbar c N^{1/3} - GNm_B^2 \right)$$

Stability requires:

$$N < N_{\max} \sim \left(\frac{\hbar c}{G m_B^2} \right)^{3/2}$$

which corresponds to

$$M_{\text{Ch}} \approx 1.4 M_{\odot}$$

In-class question

Why is the maximum mass of neutron stars lower than this simple estimate?

► Solution

Consequences for Black Holes

Stars above the Chandrasekhar mass cannot end as white dwarfs. Depending on mass loss and collapse physics, they form:

- Neutron stars
- Black holes

The collapse releases enough energy to power a core-collapse supernova.

In-class Exercise: Estimate of a White Dwarf Radius

Task:

1. Assume a white dwarf of mass $M = 1 M_{\odot}$ composed of fully ionized carbon ($m_B \approx 2m_p$ per electron).
2. Using the nonrelativistic scaling

$$R \sim \frac{\hbar^2}{G m_B^2 M^{1/3} m_e}$$

estimate the radius R in cm using Python.

3. Compare your estimate to the Sun's radius ($R_{\odot} \approx 7 \times 10^{10}$ cm).

► Solution

"Almost Perfect" Equations of State

In real gases, interactions have to be taken into account that modify the “perfect” results given above. In addition, a stellar equation of state might consist of many components with radiation, Maxwell–Boltzmann, and degenerate gases competing in importance.

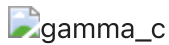
A measure of the interaction energy between two ions is the Coulomb potential

$$\phi = \frac{Z^2 e^2}{a}$$

where a is the typical separation between the two ions. Similar to past estimates, we suggest that these effects become important when they are comparable to the thermal energy kT . This allows us to define the ratio,

$$\Gamma_c = \frac{Z^2 e^2}{akT}$$

Plotting the different regimes we find:



A composite showing how the p – T plane is broken up into regions dominated by pressure ionization, degeneracy, radiation, ideal gas, crystallization, and ionization-recombination. The gas is assumed to be pure hydrogen. From HKT 3.9.

Roughly, for $\Gamma_v > 1$ Coulomb effects may be important. For values of $\Gamma_v > 170$ and appropriate densities such as in white dwarfs, Coulomb effects overwhelm those of thermal agitation and the gas settles down into a crystal. Also noted is the boundary at which the Saha equation assumptions terminate – *pressure ionization*.

Adiabatic Exponents and Other Derivatives

Keeping the Composition Fixed

Specific Heats

$$c_\alpha = \left(\frac{dQ}{dT} \right)_\alpha$$

where α is kept fixed as T changes.

Units

Q - (erg g⁻¹)

c_α - (erg g⁻¹K⁻¹)

From HKT Eq. 1. we have:

$$dQ = dE + PdV_\rho = dE + Pd\left(\frac{1}{\rho}\right) = dE - \frac{P}{\rho^2}d\rho$$

where $V_\rho = 1/\rho$ is the specific volume.

This leads us to

$$c_{V_\rho} = \left(\frac{dQ}{dT}\right)_\rho = \left(\frac{\partial E}{\partial T}\right)_\rho \text{ (erg g}^{-1}\text{K}^{-1}\text{)}$$

for an ideal monotomic gas we have $E = 3N_A kT/2\mu$ such that $c_{V_\rho} = 3N_A k/2\mu$ or $E = c_{V_\rho} T$.

To find the specific heat at constant pressure we start with the relation:

$$c_P - c_{V_\rho} = -T \left(\frac{\partial P}{\partial T}\right)_{\rho \text{ or } V_\rho}^2 \left(\frac{\partial P}{\partial V_\rho}\right)_T^{-1}$$

using the power-law expression for pressure from HKT 1.67, $P = P_0 \rho^{\chi_\rho} T^{\chi_T}$.

This gives the following definitions for the constants χ_T and χ_ρ :

$$\chi_T = \left(\frac{\partial \ln P}{\partial \ln T}\right)_{\rho \text{ or } V_\rho} = \frac{T}{P} \left(\frac{\partial P}{\partial T}\right)_{\rho \text{ or } V_\rho}$$

and

$$\chi_\rho = \left(\frac{\partial \ln P}{\partial \ln \rho}\right)_T = -\left(\frac{\partial \ln P}{\partial \ln V_\rho}\right)_T = \frac{\rho}{P} \left(\frac{\partial P}{\partial \rho}\right)_T = -\frac{1}{\rho P} \left(\frac{\partial P}{\partial V_\rho}\right)_T$$

Finally, combining this all together we find a reduced expression:

$$c_P - c_{V_\rho} = \frac{P}{\rho T} \frac{\chi_T^2}{\chi_\rho}$$

For an **Ideal Gas**, $\chi_T = \chi_\rho = 1$ giving

$$c_P - c_{V_\rho} = \frac{N_A k}{\mu}$$

and finally $c_P = 5N_A k/2\mu$.

We also define a new variable

$$\gamma = \frac{c_P}{c_{V_\rho}} \equiv 1 + \frac{P}{\rho T c_{V_\rho}} \frac{\chi_T^2}{\chi_\rho}$$

Adiabatic Exponents

The dimensionless adiabatic exponents, the Γ 's measure the thermodynamic response of the system to adiabatic changes and will be used extensively. As in Chapter 1, the subscript "ad" means that the indicated partials are to be evaluated at constant entropy.

$$\begin{aligned}\Gamma_1 &= \left(\frac{\partial \ln P}{\partial \ln \rho} \right)_{\text{ad}} = - \left(\frac{\partial \ln P}{\partial \ln V_\rho} \right)_{\text{ad}} \\ \frac{\Gamma_2}{\Gamma_2 - 1} &= \left(\frac{\partial \ln P}{\partial \ln T} \right)_{\text{ad}} = \frac{1}{\nabla_{\text{ad}}} \\ \Gamma_3 - 1 &= \left(\frac{\partial \ln T}{\partial \ln \rho} \right)_{\text{ad}} = - \left(\frac{\partial \ln T}{\partial \ln V_\rho} \right)_{\text{ad}}\end{aligned}$$

Finally, we have

$$\frac{\Gamma_3 - 1}{\Gamma_1} = \frac{\Gamma_2 - 1}{\Gamma_2} = \nabla_{\text{ad}}.$$

Now, we can write some more useful versions of these exponents and relating to the specific heats:

$$\begin{aligned}\Gamma_3 - 1 &= \frac{P}{\rho T} \frac{\chi_T}{c_{V_\rho}} = \frac{1}{\rho} \left(\frac{\partial P}{\partial E} \right)_\rho \\ \Gamma_1 &= \chi_T (\Gamma_3 - 1) + \chi_\rho = \frac{\chi_\rho}{1 - \chi_T \nabla_{\text{ad}}} \\ \frac{\Gamma_2}{\Gamma_2 - 1} &= \nabla_{\text{ad}}^{-1} = c_P \frac{\rho T}{P} \frac{\chi_\rho}{\chi_T} = \frac{\chi_\rho}{\Gamma_3 - 1} + \chi_T\end{aligned}$$

and lastly, γ

$$\gamma = \frac{c_P}{c_{V_\rho}} = \frac{\Gamma_1}{\chi_\rho} = 1 + \frac{\chi_T}{\chi_\rho} (\Gamma_3 - 1) = \frac{\Gamma_3 - 1}{\chi_\rho} \frac{1}{\nabla_{\text{ad}}}$$

Mixtures of Ideal Gases and Radiation

For stellar models, an equation of state that is a mixture of ideal gas and radiation can suffice, allowing us to write the pressure as:

$$P_{\text{total}} = \frac{\rho N_{\text{A}} k T}{\mu} + \frac{a T^4}{3} = P_g + P_{\text{rad}}.$$

and the energy in a similar fashion

$$E_{\text{total}} = \frac{3 N_{\text{A}} k T}{2 \mu} + \frac{a T^4}{\rho} = P_g + P_{\text{rad}}.$$

Be sure to keep track of V and V_ρ when using any of the above equations!

In-Class Assignment

Learning Objectives

- explore EOS regimes
- label various EOS regimes
- plot stellar models on EOS regimes
- compare density profiles estimates to stellar models

Physical constants, all in CGS

EOS regimes

We can look at the ρ - T plane to understand where each of the components of the EOS dominate.

Some composition info. We'll specify

$$\mu_e = \left[\sum_k \frac{Z_k X_k}{A_k} \right]^{-1}$$
$$\mu_I = \left[\sum_k \frac{X_k}{A_k} \right]^{-1}$$

and μ such that

$$\frac{1}{\mu} = \frac{1}{\mu_e} + \frac{1}{\mu_I}$$

The model files are again:

- $1M_\odot$: [M1_default_trimmed_history.data](#)
- $8M_\odot$: [M8_basic_co_trimmed_history.data](#)
- $15M_\odot$: [M15_aprox21_trimmed_history.data](#)

Degeneracy vs. Ideal gas

We find the (dimensionless) Fermi momentum ($x_F = p_F/(mc)$) via

$$n_e = \frac{\rho}{\mu_e m_u} = \int_0^{x_F} x^2 dx$$

where we assume complete degeneracy (so the distribution function is just a step-function). This gives

$$x_F = \left(\frac{\rho}{\mu_e B} \right)^{1/3}$$

where the constant B is:

$$B = \frac{8\pi}{3} m_u \left(\frac{m_e c}{h} \right)^3$$

We then define the transition to degeneracy by comparing to the thermal energy, kT

$$E_F = m_e c^2 \left(\sqrt{1 + x_F^2} - 1 \right) \sim kT$$

Here's a function that computes this line

Ideal gas vs. radiation

A simple way to determine whether an ideal case or radiation dominates is to set their energy densities equal. At the temperatures where we expect this to occur, we will assume that the electrons are not degenerate, so we will use the full μ for the composition in the ideal gas instead of μ_I .

Our line is:

$$\frac{\rho kT}{\mu m_u} \sim \frac{1}{3} a T^4$$

Here's a function that computes this

Finally, at low temperatures and high densities, the ions can crystalize. The comparison where is the thermal energy to the Coulomb energy:

$$\Gamma = \frac{kT}{Z^2 e^2 / a}$$

where we take $\Gamma = 170$ as the point where crystalization sets in.

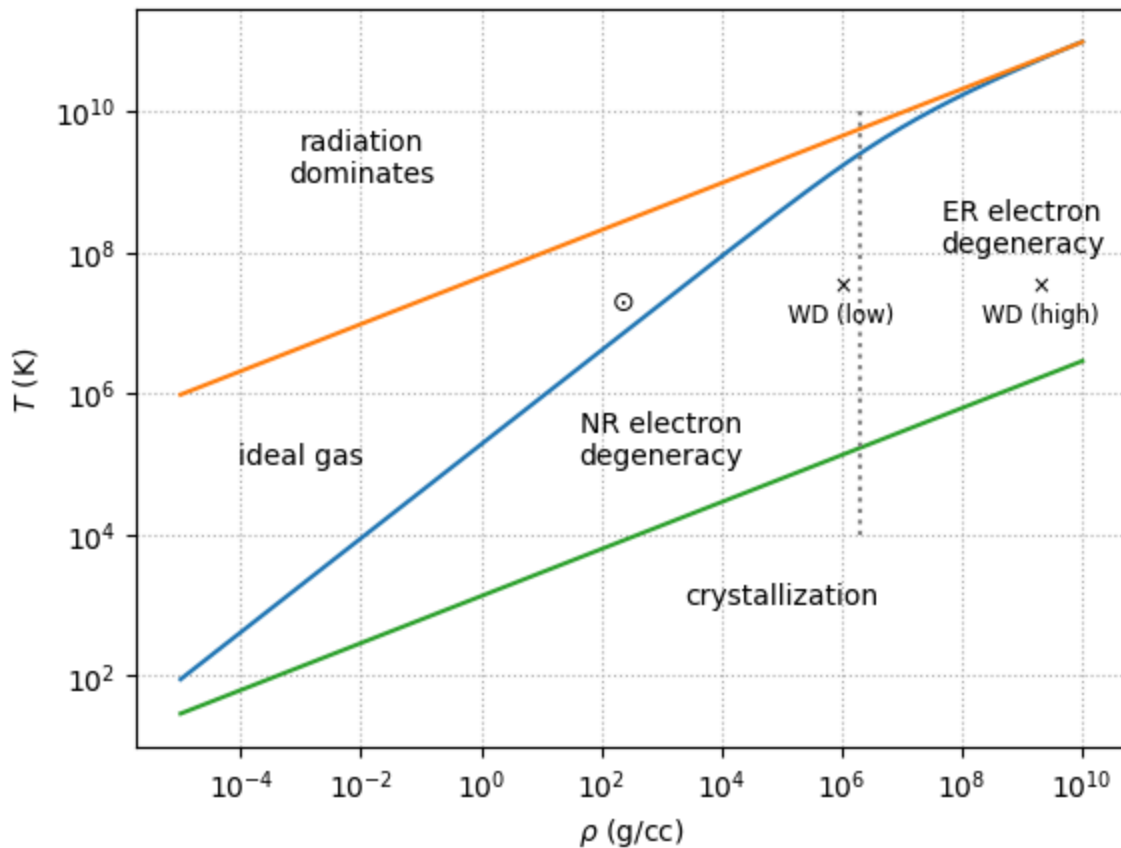
The separation between ions, a is found via:

$$V = n_I^{-1} = \frac{\mu_I m_u}{\rho} = \frac{4\pi}{3} a^3$$

Putting this together, the expression for the boundary of crystallization is computed below

Plotting the boundaries

Plot and label the three lines defined above. Label the *radiation dominated*, *ideal gas*, the electron degenerate regime (including a line denoting the density from non-relativistic to relativistic regimes), and lastly, label the crystallization regime.



Plotting the boundaries + MESA

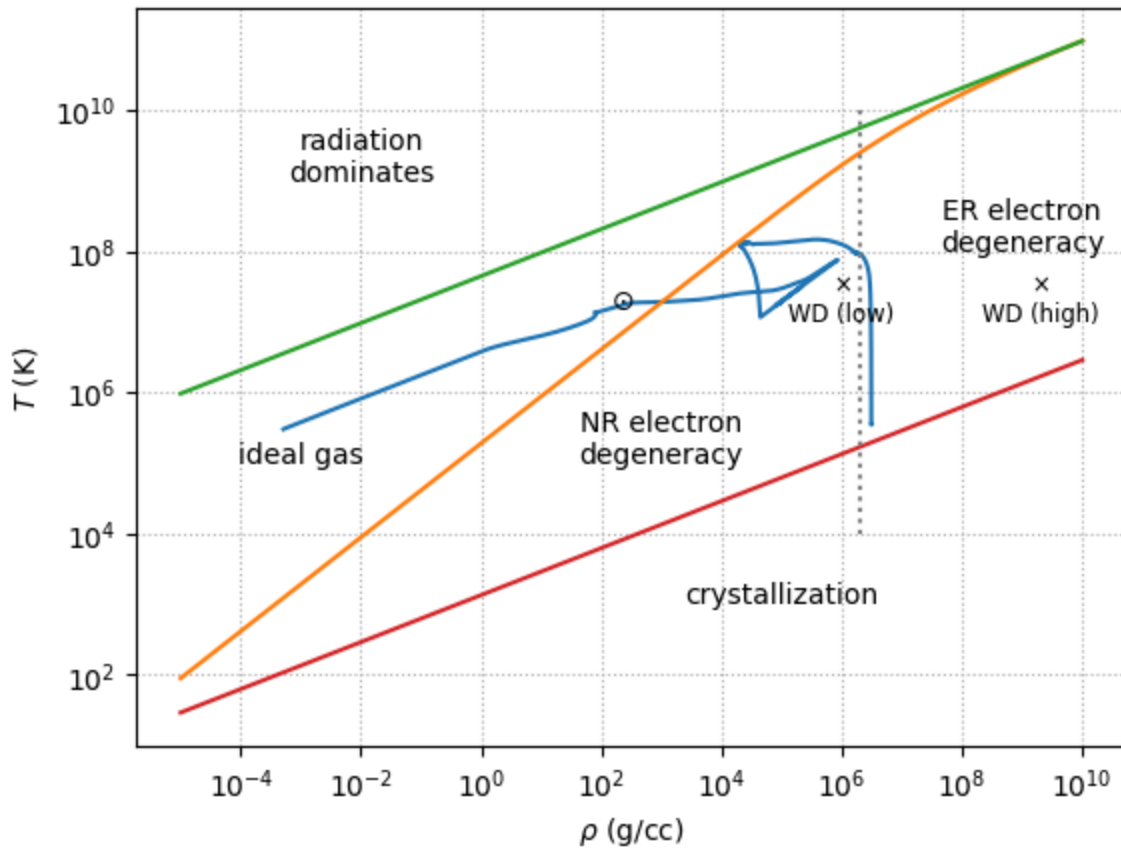
Here, produce the same plot but overlay the results of the 1M MESA stellar model history data from ICA4.

To plot the results all on the same plot, you will need to match the size of rho to the number of points for MESA.

One strategy would be to

1. load the MESA data using pandas,
2. find the number of cells,
3. then create a new rho array for use with the above functions that matches the size of the history central ρ array.

```
Index(['model_number', 'star_age', 'star_mass', 'log_L', 'log_R', 'log_Teff',  
      'log_center_T', 'log_center_Rho', 'log_center_P', 'center_h1',  
      'center_he3', 'center_he4', 'center_c12', 'center_n14', 'center_o16',  
      'center_ne20', 'center_mg24', 'center_si28', 'center_s32',  
      'center_ar36', 'center_ca40', 'center_ti44', 'center_cr48',  
      'center_fe52', 'center_fe54', 'center_fe56', 'center_ni56',  
      'center_degeneracy', 'center_ye', 'center_entropy',  
      'compactness_parameter', 'dynamic_timescale', 'kh_timescale',  
      'nuc_timescale', 'pp', 'cno', 'tri_alfa', 'log_LH', 'log_LHe', 'log_LZ',  
      'log_Lneu', 'he_core_mass', 'c_core_mass', 'o_core_mass',  
      'si_core_mass', 'fe_core_mass', 'he_core_radius', 'c_core_radius',  
      'o_core_radius', 'si_core_radius', 'fe_core_radius',  
      'max_abs_v_velocity', 'surf_avg_omega_div_omega_crit',  
      'log_total_angular_momentum', 'surf_avg_omega', 'surf_avg_v_rot',  
      'star_mdot'],  
      dtype='object')
```



```
Index(['mass', 'radius', 'luminosity', 'pressure', 'logRho', 'logT', 'energy',
      'entropy', 'cp', 'gamma1', 'grada', 'mu', 'free_e', 'ye', 'pgas',
      'prad', 'gradr', 'gradT', 'velocity', 'conv_vel', 'opacity', 'eps_nuc',
      'pp', 'cno', 'tri_alfa', 'eps_nuc_neu_total', 'non_nuc_neu', 'eps_grav',
      'h1', 'he3', 'he4', 'c12', 'n14', 'o16', 'ne20', 'mg24', 'si28', 's32',
      'ar36', 'ca40', 'ti44', 'cr48', 'fe52', 'fe54', 'fe56', 'ni56', 'eta',
      'log_omega', 'v_rot', 'j_rot', 'dynamo_log_B_r', 'dynamo_log_B_phi',
      'log_D_conv', 'log_D_semi', 'log_D_ovr', 'log_D_thrm'],
      dtype='object')
```

154.2910716421559

[<matplotlib.lines.Line2D at 0x13d39d890>]

